Final Presentation — CNN-recurrent connectivity hybrids for visual working memory tasks

Neuro 140/240

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Motivating question

- Delay match-to-sample (DMTS) tasks
- CNN and RNN use cases
- Our architectures

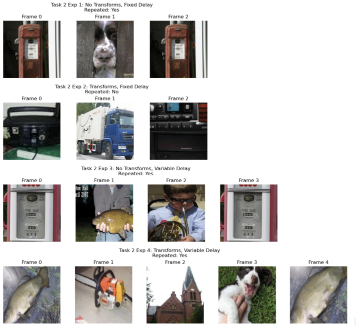


Figure: Example DMTS experiments

Methodology

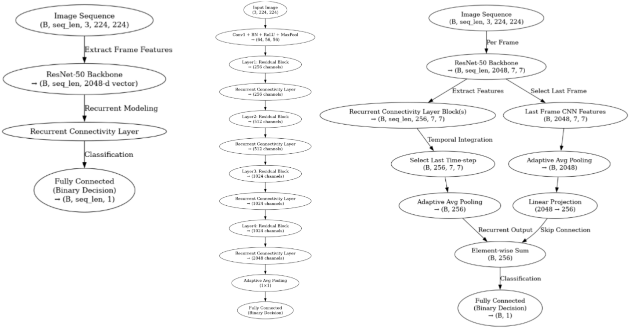


Figure: Tested CNN-recurrent connectivity hybrids

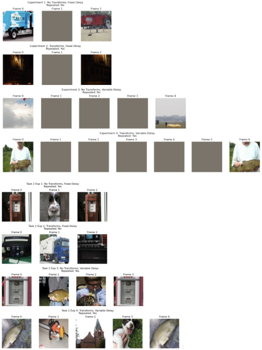
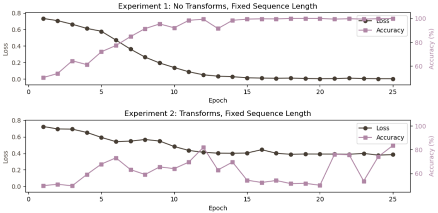


Figure: Experiments tested, ascending in complexity

Results



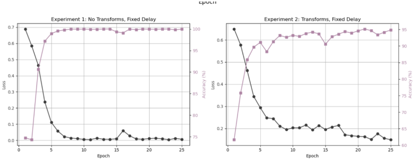


Figure: Accuracies and losses for task 2 experiments 1 and 2 (recurrent connectivity on all layers)

Results (cont.)

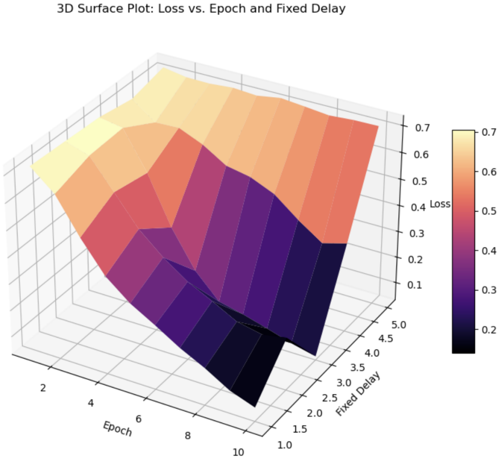


Figure: Surface plot for loss, task 1 (recurrent connectivity on all layers)

Learned representations

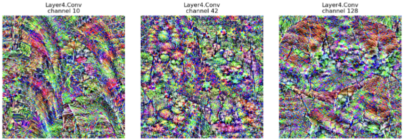


Figure: Activation maximization

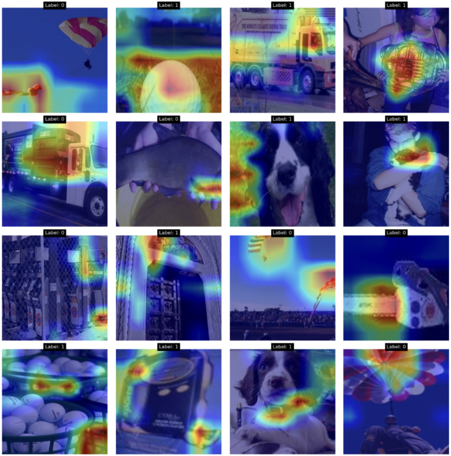


Figure: Grad-CAM

Conclusions

- Model capabilities
  - distractor vs. empty delay images, delay length, delay variability
- Biological plausibility
- Future work